HTTPX

A next-generation HTTP client for Python.

HTTPX is a fully featured HTTP client for Python 3, which provides sync and async APIs, and support for both HTTP/1.1 and HTTP/2.

Install HTTPX using pip:

```
$ pip install httpx
```

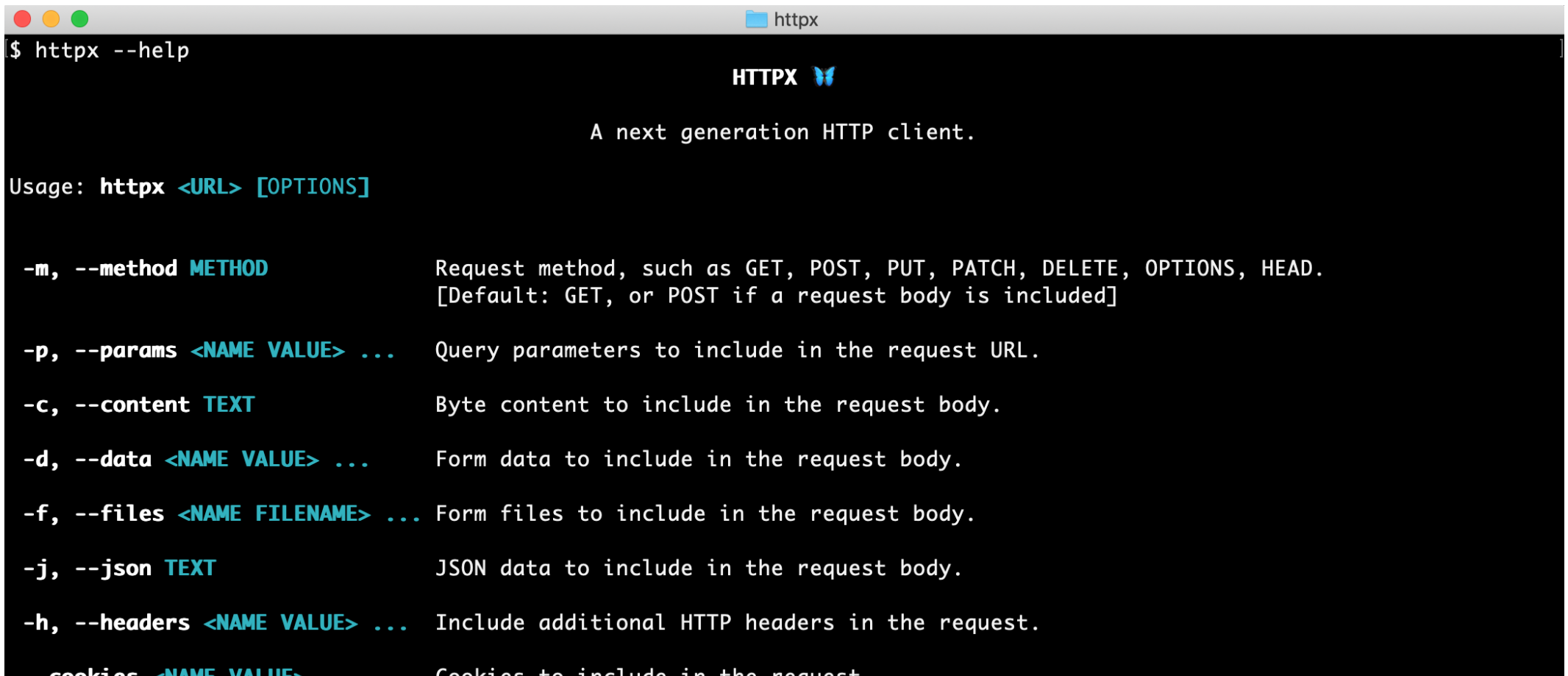
Now, let's get started:

```
>>> import httpx
>>> r = httpx.get('https://www.example.org/')
>>> r
<Response [200 OK]>
>>> r.status_code
200
>>> r.headers['content-type']
'text/html; charset=UTF-8'
>>> r.text
'<!doctype html>\n<html>\n<head>\n<title>Example Domain</title>...'
```

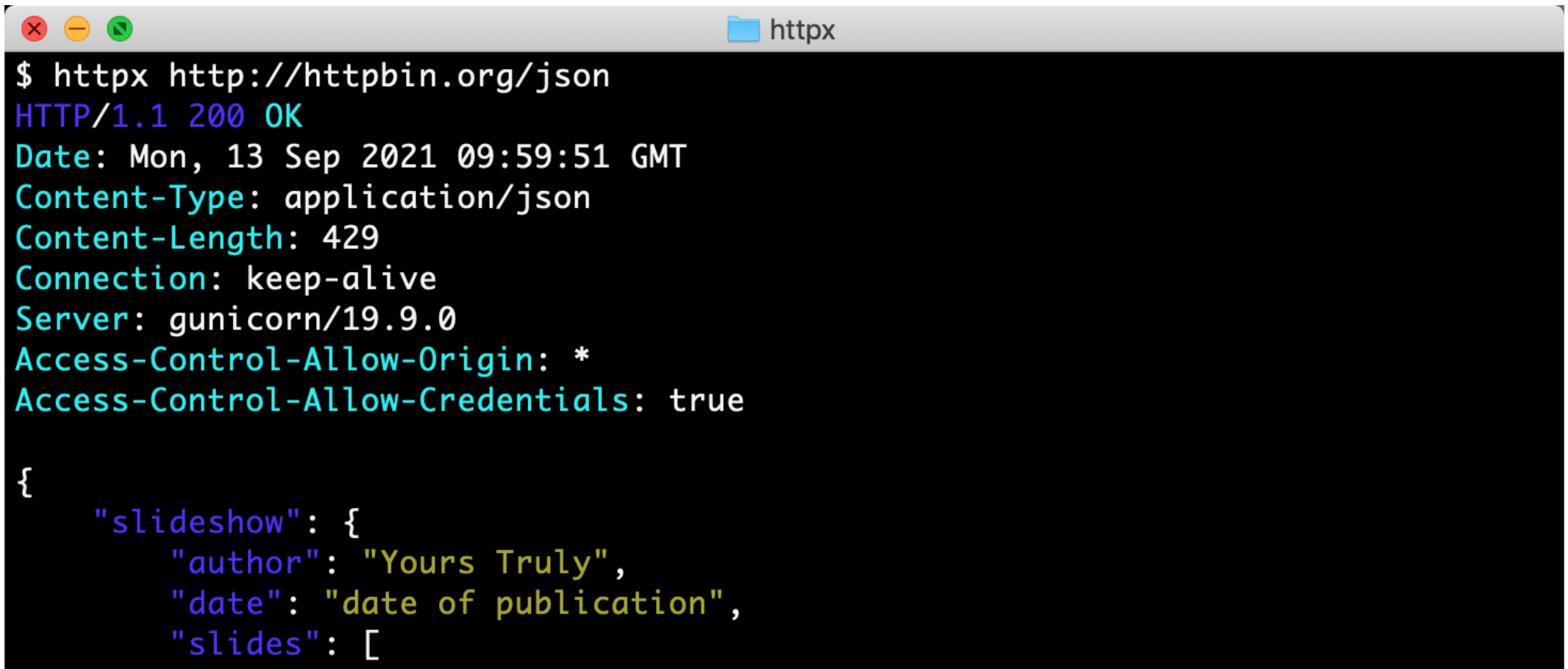
Or, using the command-line client.

```
# The command line client is an optional dependency.
$ pip install 'httpx[cli]'
```

Which now allows us to use HTTPX directly from the command-line...



Sending a request...



Features

HTTPX builds on the well-established usability of `requests`, and gives you:

- A broadly requests-compatible API.
- Standard synchronous interface, but with async support if you need it.
- HTTP/1.1 and HTTP/2 support.
- Ability to make requests directly to WSGI applications or ASGI applications.
- Strict timeouts everywhere.
- Fully type annotated.
- 100% test coverage.

Plus all the standard features of `requests`...

- International Domains and URLs
- Keep-Alive & Connection Pooling
- Sessions with Cookie Persistence
- Browser-style SSL Verification
- Basic/Digest Authentication
- Elegant Key/Value Cookies
- Automatic Decompression
- Automatic Content Decoding

- Unicode Response Bodies
- Multipart File Uploads
- HTTP(S) Proxy Support
- Connection Timeouts
- Streaming Downloads
- .netrc Support
- Chunked Requests

Documentation

For a run-through of all the basics, head over to the QuickStart.

For more advanced topics, see the **Advanced** section, the async support section, or the HTTP/2 section.

The Developer Interface provides a comprehensive API reference.

To find out about tools that integrate with HTTPX, see Third Party Packages.

Dependencies

The HTTPX project relies on these excellent libraries:

- `httpcore` - The underlying transport implementation for `httpx`.
- `h11` - HTTP/1.1 support.
- `certifi` - SSL certificates.
- `idna` - Internationalized domain name support.
- `sniffio` - Async library autodetection.

As well as these optional installs:

- `h2` - HTTP/2 support. *(Optional, with `httpx[http2]`)*
- `socksio` - SOCKS proxy support. *(Optional, with `httpx[socks]`)*
- `rich` - Rich terminal support. *(Optional, with `httpx[cli]`)*
- `click` - Command line client support. *(Optional, with `httpx[cli]`)*
- `brotli` or `brotlicffi` - Decoding for "brotli" compressed responses. *(Optional, with `httpx[brotli]`)*
- `zstandard` - Decoding for "zstd" compressed responses. *(Optional, with `httpx[zstd]`)*

A huge amount of credit is due to `requests` for the API layout that much of this work follows, as well as to `urllib3` for plenty of design inspiration around the lower-level networking details.

Installation

Install with pip:

```
$ pip install httpx
```

Or, to include the optional HTTP/2 support, use:

```
$ pip install httpx[http2]
```

To include the optional brotli and zstandard decoders support, use:

```
$ pip install httpx[brotli,zstd]
```

HTTPX requires Python 3.9+